FIREREASON: A FIRE-SPECIFIC REMOTE SENSING VISION-LANGUAGE DATASET AND FRAMEWORK FOR POST-WILDFIRE RESPONSE

Yuhang Yan and Qingyu Li

***Abstract*—Post-wildfire response requires remote sensing systems that move beyond burned-area mapping to answer structured questions about damage, accessibility, and response priorities. This paper presents Fire-ResponseQA, a fire-specific remote sensing visual question answering dataset, and FireResponder, a vision-language model (VLM) framework for structured post-fire assessment. Fire-ResponseQA contains 24,875 question-answer pairs organized into five response levels: Area & Count, Compactness & Shape, Proximity & Buffer, Graph & Connectivity, and Optimization & Cost. FireResponder adapts a vision-language backbone through supervised parameter-efficient fine-tuning and produces normalized outputs for multiple-choice, yes/no, and numerical questions. On Fire-ResponseQA, the FireResponder-7B implementation achieves a 65.09 macro average, outperforming the strongest general baseline at 24.68. These results show that fire-specific instruction tuning improves response-oriented remote sensing reasoning.**

***Index Terms*—fire disaster response, remote sensing visual question answering, vision-language model, damage assessment, accessibility reasoning.**

I. INTRODUCTION

Fire disasters rapidly reshape natural and built environments, leaving burned vegetation, damaged buildings, blocked roads, and altered accessibility conditions that must be interpreted under time pressure. Remote sensing has become a central source of post-fire evidence, supporting burned-area mapping, burn-severity assessment, and large-scale fire impact analysis from remote sensing imagery [1–3]. These observations provide spatially consistent evidence for fire assessment. Emergency response, however, also requires object-level and relation-level answers, such as which structures are affected, whether access routes remain usable, and which recovery actions should be prioritized.

Traditional remote sensing pipelines usually formulate fire assessment as fixed mapping or classification tasks, including burned-area delineation, burn-severity estimation, and damage categorization. Such outputs are valuable for scientific analysis and operational mapping, but they do not directly support flexible natural-language queries from emergency analysts. A response workflow therefore needs to integrate visual recognition, numerical estimation, spatial relation reasoning, road-network interpretation, and cost-aware decision support within the same scene.

Recent disaster datasets and remote sensing vision-language models (VLMs) provide a path toward this type of interactive assessment. Building-damage datasets such as xBD [4] and multimodal disaster datasets such as BRIGHT [5] and DisasterM3 [6] have expanded disaster understanding from image-level recognition to structured damage assessment. At the same time, remote sensing visual question answering (VQA) and VLM studies show that natural-language questions can be connected with geospatial visual evidence for recognition, counting, grounding, and reasoning [7–11].

Despite this progress, existing resources remain only partially aligned with post-fire response. Fire-specific remote sensing studies mainly emphasize mapped extent, severity, or damage categories, while general disaster VLM datasets cover multiple hazards without concentrating on fire-specific access, buffer, graph, and cost reasoning. They therefore provide limited supervision for response questions that ask not only what is damaged, but also how damage affects accessibility and restoration decisions.

To address this need, this work focuses on fire-oriented remote sensing VQA for post-fire response. We organize fire disaster assessment as a hierarchy of question-answer tasks and develop a response model that converts remote sensing observations into structured assessment outputs. The contributions are as follows:

- We construct Fire-ResponseQA, a fire-specific remote sensing vision-language dataset for post-fire damage assessment and emergency response. It contains 24,875 question-answer pairs covering five levels of assessment: Area & Count, Compactness & Shape, Proximity & Buffer, Graph & Connectivity, and Optimization & Cost.
- We propose FireResponder, a vision-language fire response framework that produces structured assessment outputs for fire-affected regions. The framework supports damage quantification, spatial relationship reasoning, road-network accessibility analysis, and response-cost estimation.

Y. Yan and Q. Li are with the School of Science and Engineering, The Chinese University of Hong Kong, Shenzhen, China. Corresponding author: Qingyu Li (liqingyu@cuhk.edu.cn). E-mail: yuhangyan@link.cuhk.edu.cn; liqingyu@cuhk.edu.cn.

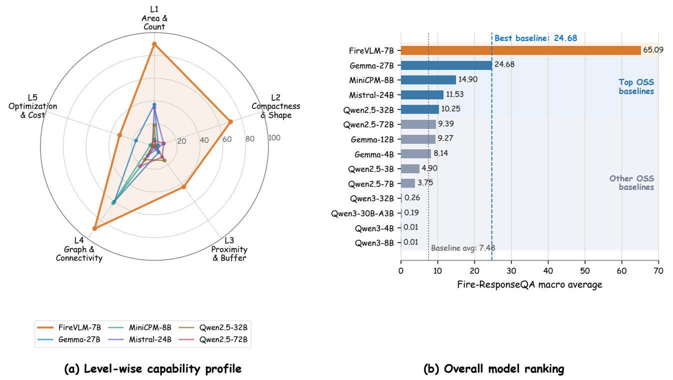

Fig. 1. Performance overview on Fire-ResponseQA. (a) Level-wise capability profiles across L1–L5 response tasks. (b) Overall macro-average ranking, with FireResponder-7B compared against the strongest general baseline and the baseline average.

II. Related Work

Remote sensing has long supported disaster and fire assessment through change detection, semantic mapping, object-level damage analysis, and post-event monitoring. For fire assessment, FireCCI provides long-term global burned-area observations from MODIS reflectance and thermal anomaly data [1], while MTBS provides Landsat-based burned-area and burn-severity maps [2]. Recent deep segmentation methods further estimate burn severity from remote sensing imagery at scale [3]. In the broader disaster domain, xBD and BRIGHT provide building-damage benchmarks from high-resolution optical and multimodal observations [4, 5]. Together, these studies establish foundations for burned-area mapping, severity estimation, and structural damage assessment.

Vision-language learning extends remote sensing interpretation from fixed visual labels to language-guided querying and reasoning. RSVQA formulates remote sensing image understanding as visual question answering [7], and EarthVQA emphasizes relational reasoning over Earth observation scenes [8]. Recent remote sensing VLMs such as GeoChat, RSGPT, and EarthGPT explore grounded dialogue, instruction following, and multisensor image comprehension [9–11]. For disaster-oriented VLMs, DisasterM3 introduces multi-hazard vision-language supervision for damage assessment and response [6], while ForestFireVLM studies VLM-based wildfire scene understanding from UAV imagery [12]. This work frames language-guided remote sensing as a useful paradigm for connecting visual evidence with disaster semantics, spatial reasoning, and response-oriented descriptions.

III. Methodology

FireResponder converts fire-related remote sensing evidence into structured response-oriented answers. It contains three components: construction of Fire-ResponseQA, parameter-efficient adaptation of a vision-language backbone, and a structured answer protocol for level-wise assessment. Figure 2 summarizes the full workflow.

A. Fire-ResponseQA Construction

Fire-ResponseQA is organized from fire-related remote sensing scenes and contains 24,875 question-answer pairs. Each sample includes image evidence, a natural-language question, a reference answer, an answer type, and a task level. When available, the visual input uses paired pre-disaster and post-disaster imagery; otherwise, the sample uses the available post-disaster optical or synthetic aperture radar (SAR) observation.

The questions are grouped into five response-oriented levels: Area & Count, Compactness & Shape, Proximity & Buffer, Graph & Connectivity, and Optimization & Cost. These levels cover basic measurement, geometric characterization, spatial relation analysis, accessibility reasoning, and response-cost estimation. The answers are normalized into three formats: multiple choice, yes/no, and numerical responses.

B. FireResponder Model Adaptation

FireResponder is implemented with Qwen2.5-VL-7B-Instruct as the vision-language backbone. We adapt the model using supervised fine-tuning with low-rank adaptation (LoRA), which updates lightweight language-side modules while preserving the pretrained visual perception components. The vision tower and multimodal projector are frozen, and LoRA is applied to the attention and multilayer perceptron (MLP) projections of the language model.

The key training settings are intentionally compact: LoRA rank is 16, LoRA alpha is 32, dropout is 0.05, the learning rate is 1×10^{-4}, and training runs for 3 epochs with bfloat16 precision. The maximum image size is limited to 65,536 pixels and the text cutoff length is 2,048 tokens. After fine-tuning, the adapter is exported with the backbone to obtain the FireResponder-7B implementation used in Table I.

C. Structured Answer Protocol and Scoring

To make fire-response outputs directly comparable, every model is instructed to place the final response inside a single <answer>...</answer> block. Multiple-choice questions require one of A, B, C, or D. Yes/no questions require either Yes or No, and numerical questions require a JSON-style numeric answer.

For evaluation, multiple-choice and yes/no questions are scored by exact accuracy. Numerical questions are evaluated with relative-error tolerance, and the main table uses the 20% relative-accuracy criterion as the primary numerical score. Level scores are computed by averaging the primary score within each L1–L5 group, and the reported average is the macro average over the five populated levels.

This protocol separates answer extraction from score computation. The parser first identifies the normalized answer inside the tagged output, and the evaluator then applies the rule associated with the corresponding answer type. This design reduces the influence of verbose explanations or minor formatting differences, which is important when comparing general VLMs with a fire-specific adapted model. Using a macro average over L1–L5 prevents the large L1 and L4

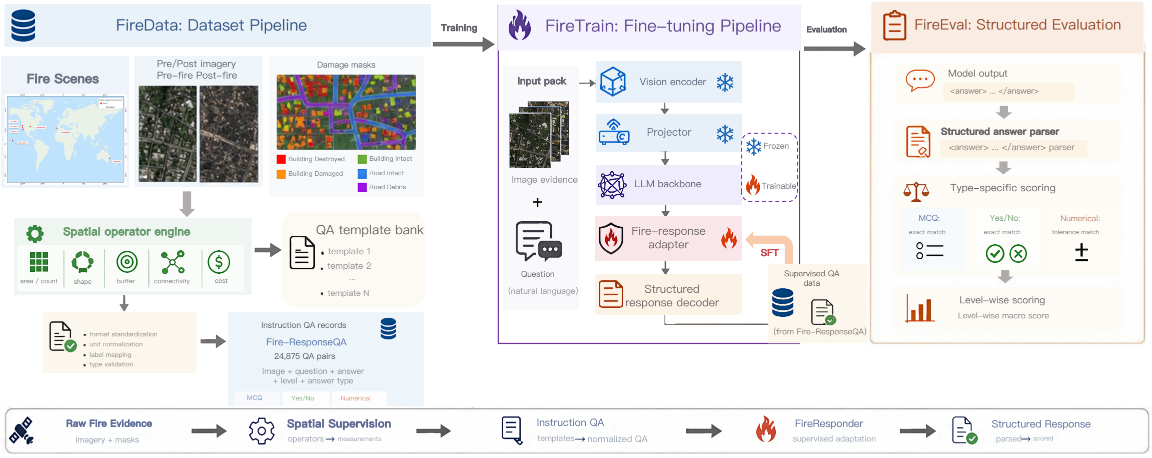

Fig. 2. Overview of FireResponder. FireData builds Fire-ResponseQA, FireTrain performs supervised fire-response adaptation, and FireEval parses structured outputs for type-specific and level-wise scoring.

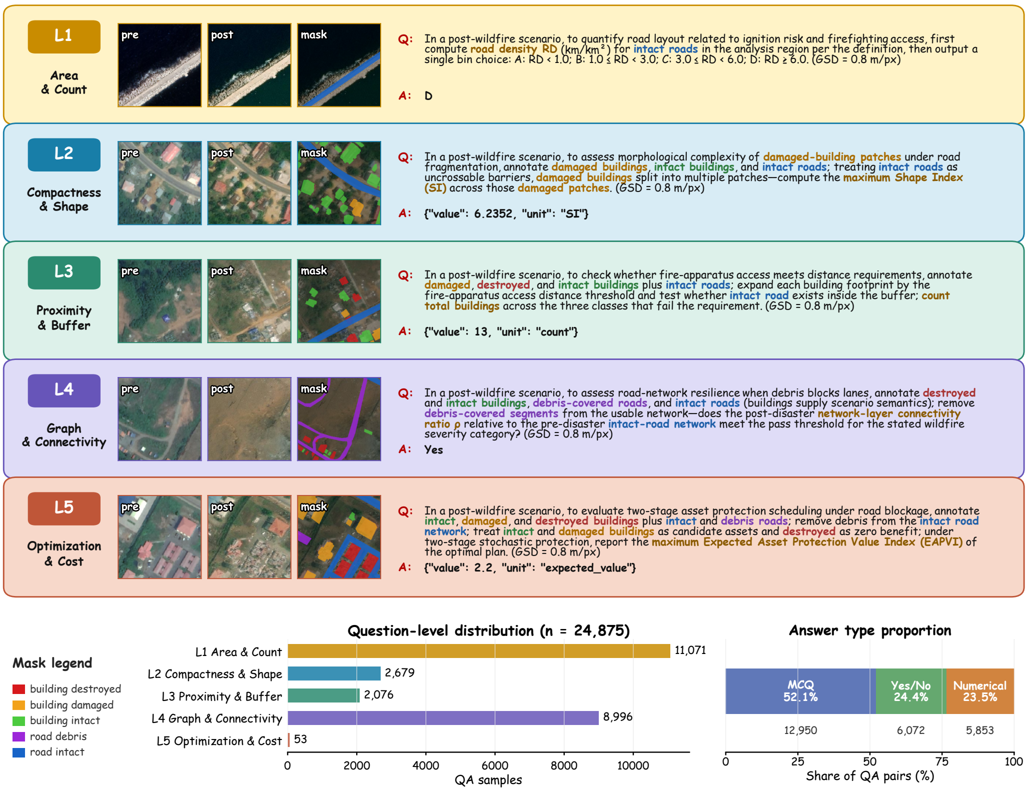

Fig. 3. Overview of Fire-ResponseQA. The dataset pairs fire-related imagery and masks with QA supervision across five response levels; the lower panels summarize level and answer-type distributions.

groups from dominating the final score. It also makes the evaluation sensitive to rare but response-critical tasks, such as proximity and optimization reasoning.

IV. Experiments

Table I reports evaluation on Fire-ResponseQA with level-wise scoring. All models use the same visual inputs, question prompts, and structured answer parser. Multiple-choice (MCQ) and yes/no samples are scored by exact accuracy, while numerical samples use 20% relative accuracy. The final average is the macro average over L1–L5. FireResponder-7B reaches 65.09, outperforming the strongest general VLM baseline, Gemma-3-27B-it, at 24.68. The gains are largest on L1 Area & Count and L4 Graph & Connectivity, while L3 and L5 remain harder because they require proximity judgment and response-planning reasoning. The comparison indicates that model scale alone is not sufficient for fire-response reasoning. Several large or XL baselines achieve non-trivial scores on individual levels, but their macro averages remain low because they fail across answer types. In contrast, FireResponder-7B benefits from fire-specific instruction tuning and the structured answer protocol. The remaining errors are concentrated in L3 and L5, where the model must combine visual evidence with buffer-based proximity judgment or optimization-style cost reasoning.

TABLE I
PERFORMANCE COMPARISON ON THE FIRE DISASTER TYPE ACROSS FIVE TASK LEVELS. ALL LISTED MODELS ARE AVAILABLE THROUGH OLLAMA.

Model	**L1**	**L2**	**L3**	**L4**	**L5**	**Avg.**
Open-source VLMs (Base Size)						
Qwen2.5-VL-3B-Instruct [13]	2.15	1.38	4.24	14.85	1.89	4.90
Qwen3-VL-4B-Instruct [14]	0.07	0.00	0.00	0.00	0.00	0.01
Gemma-3-4B-it [15]	9.20	0.63	0.10	30.77	0.00	8.14
Qwen2.5-VL-7B-Instruct [13]	1.84	2.02	1.83	13.04	0.00	3.75
MiniCPM-V-2.6-8B [16]	6.41	2.20	0.48	61.62	3.77	14.90
Qwen3-VL-8B-Instruct [14]	0.05	0.00	0.00	0.00	0.00	0.01
Gemma-3-12B-it [15]	29.47	1.46	0.34	15.06	0.00	9.27
Open-source VLMs (Large Size)						
Mistral-Small-3.2-24B-Instruct-2506 [17]	34.10	8.51	3.08	10.06	1.89	11.53
Gemma-3-27B-it [15]	36.88	3.47	6.26	59.83	16.98	24.68
Qwen3-VL-30B-A3B-Instruct [14]	0.94	0.00	0.00	0.00	0.00	0.19
Open-source VLMs (XL Size)						
Qwen2.5-VL-32B-Instruct [13]	18.70	1.31	15.17	14.18	1.89	10.25
Qwen3-VL-32B-Instruct [14]	1.32	0.00	0.00	0.00	0.00	0.26
Qwen2.5-VL-72B-Instruct [13]	5.27	8.59	11.51	21.61	0.00	9.39
Fine-tuned VLMs						
FireResponder-7B (Ours)	**90.08**	**70.32**	**43.79**	**89.16**	**32.08**	**65.09**

V. CONCLUSION

This paper introduced Fire-ResponseQA and FireResponder for fire-oriented remote sensing visual question answering. Fire-ResponseQA organizes 24,875 QA pairs into five response levels, turning post-fire imagery into structured tasks for damage, accessibility, and response planning. FireResponder adapts a vision-language backbone to produce normalized answers for multiple-choice, yes/no, and numerical fire-response questions. Results with FireResponder-7B show the benefit of fire-specific adaptation and identify proximity and optimization reasoning as key challenges for future fire-response VLMs.

REFERENCES

[1] E. Chuvieco, J. Lizundia-Loiola, M. L. Pettinari, R. Ramo, M. Padilla, K. Tansey, F. Mouillot, P. Laurent, T. Storm, A. Heil, and S. Plummer, "Generation and analysis of a new global burned area product based on modis 250 m reflectance bands and thermal anomalies," *Earth System Science Data*, vol. 10, no. 4, pp. 2015–2031, 2018.

[2] J. J. Picotte, K. Bhattarai, D. Howard, J. Lecker, J. Epting, B. Quayle, N. Benson, and K. Nelson, "Changes to the monitoring trends in burn severity program mapping production procedures and data products," *Fire Ecology*, vol. 16, no. 1, p. 16, 2020.

[3] X. Hu, P. Zhang, and Y. Ban, "Large-scale burn severity mapping in multispectral imagery using deep semantic segmentation models," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 196, pp. 228–240, 2023.

[4] R. Gupta, B. Goodman, N. Patel, R. Hosfelt, S. Sajeev, E. Heim, J. Doshi, K. Lucas, H. Choset, and M. Gaston, "Creating xbd: A dataset for assessing building damage from satellite imagery," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition workshops*, 2019, pp. 10–17.

[5] H. Chen, J. Song, O. Dietrich, C. Broni-Bediako, W. Xuan, J. Wang, X. Shao, Y. Wei, J. Xia, C. Lan *et al.*, "Bright: A globally distributed multimodal building damage assessment dataset with very-high-resolution for all-weather disaster response," *Earth System Science Data*, vol. 17, no. 11, pp. 6217–6253, 2025.

[6] J. Wang, W. Xuan, H. Qi, Z. Liu, K. Liu, Y. Wu, H. Chen, J. Song, J. Xia, Z. Zheng *et al.*, "Disasterm3: A remote sensing vision-language dataset for disaster damage assessment and response," *Advances in Neural Information Processing Systems*, vol. 38, 2026.

[7] S. Lobry, D. Marcos, J. Murray, and D. Tuia, "Rsvqa: Visual question answering for remote sensing data," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 58, no. 12, pp. 8555–8566, 2020.

[8] J. Wang, Z. Zheng, Z. Chen, A. Ma, and Y. Zhong, "Earthvqa: Towards queryable earth via relational reasoning-based remote sensing visual question answering," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 38, no. 6, 2024, pp. 5481–5489.

[9] K. Kuckreja, M. S. Danish, M. Naseer, A. Das, S. Khan, and F. S. Khan, "Geochat: Grounded large vision-language model for remote sensing," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2024, pp. 27 831–27 840.

[10] Y. Hu, J. Yuan, C. Wen, X. Lu, Y. Liu, and X. Li, "Rsgpt: A remote sensing vision language model and benchmark," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 224, pp. 272–286, 2025.

[11] W. Zhang, M. Cai, T. Zhang, Y. Zhuang, and X. Mao, "Earthgpt: A universal multimodal large language model for multisensor image comprehension in remote sensing domain," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, pp. 1–20, 2024.

[12] L. Seidel, S. Gehringer, T. Raczok, S.-N. Ivens, B. Eckardt, and M. Maerz, "Advancing early wildfire detection: Integration of vision language models with unmanned aerial vehicle remote sensing for enhanced situational awareness," *Drones*, vol. 9, no. 5, p. 347, 2025.

[13] S. Bai, K. Chen, X. Liu, J. Wang, W. Ge, S. Song, K. Dang, P. Wang, S. Wang, J. Tang *et al.*, "Qwen2. 5-vl technical report. corr, abs/2502.13923, 2025. doi: 10.48550," *arXiv preprint ARXIV.2502.13923*, 2025.

[14] S. Bai, Y. Cai, R. Chen, K. Chen, X. Chen, Z. Cheng, L. Deng, W. Ding, C. Gao, C. Ge *et al.*, "Qwen3-vl technical report," *arXiv preprint arXiv:2511.21631*, 2025.

[15] Gemma Team, A. Kamath, J. Ferret, S. Pathak, N. Vieillard, R. Merhej, S. Perrin, T. Matejovicova, A. Ramé, M. Rivière, L. Rouillard, T. Mesnard, G. Cideron, J.-B. Grill, S. Ramos, E. Yvinec, M. Casbon, E. Pot, I. Penchev *et al.*, "Gemma 3 technical report," *arXiv preprint arXiv:2503.19786*, 2025. [Online]. Available: https://arxiv.org/abs/2503.19786

[16] Y. Yao, T. Yu, A. Zhang, C. Wang, J. Cui, H. Zhu, T. Cai, H. Li, W. Zhao, Z. He *et al.*, "Minicpm-v: A gpt-4v level mllm on your phone," *arXiv preprint arXiv:2408.01800*, 2024.

[17] Mistral AI, "Model card for mistral-small-3.2-24b-instruct-2506," https://huggingface.co/mistralai/Mistral-Small-3.2-24B-Instruct-2506, 2025.